From a Real Rank-One Relation to Positive-Density Torsion Transversality

A Kummer–Chebotarev and Algebraic K-Theory Progress Report on the Alaoglu–Erdős Integer-Exponent Conjecture

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New material beyond the attached *Unified Report A*

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Abstract

The Alaoglu–Erdős integer-exponent conjecture asks whether

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z} \qquad (x \in \mathbb{R}).$$

The attached *Unified Report A* already develops an unusually broad corpus: exact conditional structure, machine-checked primitive-output normalization, transcendence reductions, finite-difference and determinant methods, p-adic diagnostics, interpolation, symbolic dynamics, canonical products, and twenty-seven continuation reports. The present report deliberately does not reproduce those arguments. It imports only two exact black boxes from that corpus and develops a new finite-torsion branch.

For a putative least counterexample β, write $M = 2^\beta$ and $A = 3^\beta$, and form the homogeneous quadratic prime-exponent polynomial

$$Q_{M,A}(X) = L_M(X)X_3 - L_A(X)X_2, \qquad L_u(X) = \sum_p v_p(u)X_p.$$

The primitive-output theorem in *Unified Report A* implies the new identity

$$\operatorname{cont}(Q_{M,A}) = \gcd(d, e, |a-b|) = 1$$

for its simultaneous normalization $M = 2^a w^d$, $A = 3^b v^e$. Thus $Q_{M,A}$ remains nonzero modulo every prime. Kummer theory and Chebotarev then turn this elementary content calculation into uniform local theorems at every prime torsion level. At the quadratic level, a multiquadratic Chebotarev argument and the minimum-distance bound for degree-two Boolean polynomials force detection at density at least $1/4$ among all rational primes. For every odd prime q, a positive-density set of primes $\ell \equiv 1 \pmod q$ detects rank two in the q-th power-residue characters of $(2, 3; M, A)$. The relative detecting density is at least $(1-1/q)^2$; for cubic characters it is at least $4/9$ among $\ell \equiv 1 \pmod 3$, hence at least $2/9$ among all primes. For all but finitely many odd q, the rank-three and rank-four branches have exact relative densities $1 - 1/q$ and $1 - 1/q - (q-1)/q^3$, respectively. Under GRH for fixed-degree Kummer fields, quadratic and cubic detecting primes satisfy $\ell \ll (\log(6MA))^2$.

These are unconditional structural advances, but they do not prove the conjecture. Instead, they isolate a sharp missing bridge: the real logarithmic rank-one relation would have to force vanishing of the cubic determinant on more than $5/9$ of the primes $\ell \equiv 1 \pmod 3$, contradicting Chebotarev. A second new branch, through Milnor $K_2(\mathbb{Q})$ and tame symbols, gives an exact congruence sieve and also proves a parity obstruction: ordinary odd-primary cup products are alternating, whereas the Alaoglu–Erdős period is symmetric. This explains why a direct reciprocity proof cannot simply be read off from the real determinant.

No full proof is claimed. Every theorem, conditional theorem, computation, and open transfer target is labelled separately.

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1 Executive verdict and claim ledger

Current status

No proof of the Alaoglu–Erdős conjecture was obtained. The main advance is a new, fully rigorous Kummer–Chebotarev theorem that forces positive-density finite torsion transversality for every hypothetical least counterexample. The theorem is compatible with the conjecture being false, so it is a reduction and a source of certificates, not a contradiction by itself.

The report's claims are divided into four levels.

Level	Meaning
Proved here	Complete paper proof is supplied from standard stated inputs.
Inherited	Used as a black box from the attached *Unified Report A*; not reproved here.
Conditional	The conclusion is exact under a named hypothesis such as GRH or a proposed transfer theorem.
Computational	Reproducible finite experiment; never used as proof of an infinite statement.
Open target	A precise statement whose proof would close a remaining gap.

Principal new conclusions

Assume the conjecture fails, let β be its least nonintegral solution, and put $M = 2^\beta$, $A = 3^\beta$.

1. The quadratic prime-exponent obstruction $Q_{M,A}$ is primitive over $\mathbb{Z}$.
2. The quadratic-character determinant is nonzero at a set of rational primes of density at least $1/4$.
3. For every odd prime q, the q-th power-residue determinant is nonzero on a positive-density set of primes. The absolute density is at least $(q-1)/q^2$.
4. At $q = 3$, the detecting set has relative density at least $4/9$ inside $\ell \equiv 1 \pmod 3$, and absolute density at least $2/9$.
5. Under GRH, quadratic and cubic detecting primes exist below effective constants times $(\log(6MA))^2$; unconditionally one obtains fixed powers of $6MA$.
6. A theorem forcing cubic determinant zero on a relative density greater than $5/9$ would prove the conjecture.
7. The ordinary K_2/cup-product route sees an alternating symbol with the wrong sign symmetry. It yields congruence restrictions but not the real period relation.

The source audit used the attached unified manuscript. A structural and keyword audit found extensive treatment of real and p-adic logarithmic determinants, but no Chebotarev density theorem, power-residue character determinant, Milnor K_2, tame-symbol, or norm-residue formulation. Those are the delta developed below.

2 Minimal inherited input from *Unified Report A*

Only the following two pieces of the attachment are imported.

2.1 Least counterexample and simultaneous primitive outputs

Assume failure and let $\beta > 0$ be the least nonintegral real number with $2^\beta, 3^\beta \in \mathbb{Z}$. Set

$$M = 2^\beta, \qquad A = 3^\beta.$$

The attachment proves that β is irrational and supplies factorizations

$$M = 2^a w^d, \qquad A = 3^b v^e, \tag{2.1}$$

where

- $a, b \geq 0$, $d, e \geq 1$;
- $w > 1$ is odd and power-primitive;
- $v > 1$ is power-primitive and $3 \nmid v$;
- $a = 0$ or $b = 0$; and
- the exact affine-primitivity identity holds:

$$\gcd\bigl(d, e, |a - b|\bigr) = 1. \tag{2.2}$$

Here power-primitive means that the gcd of the positive prime valuations is one. Thus if

$$w = \prod_{p|w} p^{r_p}, \qquad v = \prod_{p|v} p^{s_p},$$

then $\gcd_p r_p = \gcd_p s_p = 1$.

2.2 The depth-two symmetric tensor has rank three or four

Let

$$\Gamma = \mathbb{Q}^\times_{>0} \cong \bigoplus_p \mathbb{Z}e_p, \qquad \nu(u) = \sum_p v_p(u) e_p.$$

The real relation

$$\log_2 M = \log_3 A = \beta$$

is equivalent to the vanishing of the period of

$$\tau_{M,A} = \nu(M) \odot e_3 - \nu(A) \odot e_2 \in \mathrm{Sym}^2(\Gamma \otimes \mathbb{Q}), \tag{2.3}$$

under $e_p \odot e_r \mapsto \log p \log r$. The attachment combines a finite rank classification with Baker's linear-independence theorem [3] to show that, for a hypothetical counterexample, $\tau_{M,A}$ has quadratic rank three or four. In the rank-four branch it is the pullback of the split hyperbolic form $z_1 z_2 - z_3 z_4$.

Nothing else from the 27-report corpus is needed for the new theorems.

3 The primitive quadratic obstruction

Introduce one variable X_p for each prime and, for $u \in \mathbb{Q}^\times_{>0}$, the finite linear form

$$L_u(X) = \sum_p v_p(u) X_p.$$

The polynomial attached to an output pair is

$$Q_{M,A}(X) = L_M(X) X_3 - L_A(X) X_2. \tag{3.1}$$

It is the diagonal evaluation of the symmetric tensor in (2.3). In particular,

$$Q_{M,A}((\log p)_p) = \log M \log 3 - \log A \log 2. \tag{3.2}$$

Thus a counterexample gives a nonzero rational quadratic polynomial having one distinguished transcendental zero.

For an integral polynomial in finitely many variables, $\operatorname{cont}(Q)$ denotes the gcd of its coefficients, taken nonnegative.

Theorem 3.1 (Exact content formula)**.** *For the simultaneous normalization* (2.1),

$$\operatorname{cont}(Q_{M,A}) = \gcd(d, e, |a-b|). \tag{3.3}$$

Consequently a least hypothetical counterexample has $\operatorname{cont}(Q_{M,A}) = 1$*.*

Proof. Write

$$W(X) = \sum_{p|w} r_p X_p, \qquad V(X) = \sum_{p|v} s_p X_p.$$

Since w is odd, W has no X_2-term; since $3 \nmid v$, V has no X_3-term. From (2.1),

$$L_M = aX_2 + dW,$$
$$L_A = bX_3 + eV,$$

and hence

$$Q_{M,A} = (a-b)X_2X_3 + dW(X)X_3 - eV(X)X_2. \tag{3.4}$$

There is no hidden cancellation in this expansion. The first displayed monomial has coefficient $a-b$. The coefficients contributed by dWX_3 have gcd d, because $\gcd_p r_p = 1$. The coefficients contributed by eVX_2 have gcd e, because $\gcd_p s_p = 1$. Even if a prime divides both w and v, the corresponding monomials are X_pX_3 and X_pX_2, so they remain distinct. If $3 \mid w$ or $2 \mid v$, the extra monomials are X_3^2 or X_2^2, again distinct. Therefore the gcd of all coefficients is precisely $\gcd(d, e, |a-b|)$. Equation (2.2) proves the final claim. ☐

Corollary 3.2 (Universal nonvanishing after reduction)**.** *For a least hypothetical counterexample, the reduction of $Q_{M,A}$ is a nonzero quadratic polynomial over $\mathbb{F}_q$ for every prime q.*

Proof. A primitive integral polynomial cannot have every coefficient divisible by q. ☐

Remark 3.3. The gcd-one output theorem in *Unified Report A* was originally a descent statement. The new point is that it is exactly the coefficient-content statement needed to keep the symmetric obstruction alive at every torsion level. This is the hinge between the real-period problem and Kummer theory.

3.1 Standard pairs are exactly the zero-polynomial pairs

Proposition 3.4 (Formal standardness criterion)**.** *For positive integers M, A, the following are equivalent:*

1. *$Q_{M,A} = 0$ over $\mathbb{Z}$;*
2. *there is an integer $n \geq 0$ such that $M = 2^n$ and $A = 3^n$.*

Proof. The reverse implication is immediate. Conversely, compare coefficients in $L_MX_3 - L_AX_2$. The coefficient of X_pX_3, for $p \notin \{2, 3\}$, is $v_p(M)$, while the coefficient of X_pX_2 is $-v_p(A)$; hence both vanish. The coefficients of X_3^2 and X_2^2 give $v_3(M) = v_2(A) = 0$. Finally the coefficient of X_2X_3 gives $v_2(M) = v_3(A) = n$. ☐

The contrast is already sharp:

$$\begin{array}{r|c} \text{integer-exponent output pair} & Q_{M,A} = 0, \\ \text{least hypothetical noninteger output pair} & \operatorname{cont}(Q_{M,A}) = 1. \end{array}$$

The next sections make this integral dichotomy visible in reductions modulo rational primes.

4 Kummer characters and the finite determinant

Fix an odd prime q. Let

$$\overline{\Gamma}_q = \mathbb{Q}^{\times}_{>0}/(\mathbb{Q}^{\times}_{>0})^q \cong \bigoplus_p \mathbb{F}_q e_p$$

and let

$$W_q = \operatorname{span}_{\mathbb{F}_q}\{[2],[3],[M],[A]\} \subset \overline{\Gamma}_q, \qquad r_q = \dim_{\mathbb{F}_q} W_q. \tag{4.1}$$

Because the classes of 2 and 3 are independent, $2 \le r_q \le 4$.

Put $K_q = \mathbb{Q}(\zeta_q)$, where ζ_q is a primitive q-th root of unity, and consider

$$\mathcal{L}_q = K_q(2^{1/q}, 3^{1/q}, M^{1/q}, A^{1/q}). \tag{4.2}$$

Lemma 4.1 (No loss of rational Kummer rank)**.** *The natural map*

$$\mathbb{Q}^{\times}/(\mathbb{Q}^{\times})^q \longrightarrow K_q^{\times}/(K_q^{\times})^q$$

is injective. In particular,

$$\operatorname{Gal}(\mathcal{L}_q/K_q) \cong \operatorname{Hom}_{\mathbb{F}_q}(W_q, \mu_q) \quad \textit{and} \quad [\mathcal{L}_q : K_q] = q^{r_q}. \tag{4.3}$$

Proof. Suppose $u \in \mathbb{Q}^{\times}$ becomes a q-th power in K_q, say $u = y^q$. Taking norms to $\mathbb{Q}$ gives

$$u^{q-1} = N_{K_q/\mathbb{Q}}(y)^q.$$

Thus the class of u in $\mathbb{Q}^{\times}/(\mathbb{Q}^{\times})^q$ is annihilated by $q-1$. It is also q-torsion, and $\gcd(q, q-1) = 1$, so the class is zero. The Kummer description then follows from the standard Kummer theorem over a field containing μ_q [6]. □

Let ℓ be a rational prime satisfying

$$\ell \equiv 1 \pmod q, \qquad \ell \nmid 6MAq.$$

Choose $\omega_\ell \in \mathbb{F}_\ell^{\times}$ of exact order q. For $u \in \mathbb{Q}^{\times}$ with $\ell \nmid u$, define the additive q-th power-residue character $\chi_{\ell,q}(u) \in \mathbb{F}_q$ by

$$u^{(\ell-1)/q} \equiv \omega_\ell^{\chi_{\ell,q}(u)} \pmod \ell. \tag{4.4}$$

It is a homomorphism $\overline{\Gamma}_q \to \mathbb{F}_q$. Replacing ω_ℓ by another generator scales every character value by one common nonzero scalar.

Definition 4.2 (Power-residue determinant)**.** Set

$$\Delta_{\ell,q}(M,A) = \chi_{\ell,q}(M)\chi_{\ell,q}(3) - \chi_{\ell,q}(A)\chi_{\ell,q}(2) \in \mathbb{F}_q. \tag{4.5}$$

Its vanishing is independent of the choice of ω_ℓ, because a change of generator multiplies $\Delta_{\ell,q}$ by a nonzero square.

Proposition 4.3 (The determinant is the quadratic obstruction evaluated at Frobenius)**.** *For every eligible ℓ,*

$$\Delta_{\ell,q}(M,A) = Q_{M,A}(\chi_{\ell,q}). \tag{4.6}$$

Moreover, up to the common scalar ambiguity just described, $\chi_{\ell,q}|_{W_q}$ is the Kummer character of Frobenius at a prime of K_q above ℓ.

Proof. The first identity follows by substituting $X_p = \chi_{\ell,q}(p)$ into (3.1) and using additivity on prime valuations. The second statement is the usual power-residue description of Frobenius in a Kummer extension. □

For a standard pair $(M,A) = (2^n, 3^n)$, both rows of the character matrix are proportional:

$$\begin{pmatrix} \chi(2) & \chi(M) \\ \chi(3) & \chi(A) \end{pmatrix} = \begin{pmatrix} \chi(2) & n\chi(2) \\ \chi(3) & n\chi(3) \end{pmatrix},$$

so $\Delta_{\ell,q} = 0$ for every q and every eligible ℓ. A least hypothetical counterexample will exhibit the opposite behavior on a positive-density set at every odd level.

5 Chebotarev forces positive-density torsion transversality

The extension $\mathcal{L}_q/\mathbb{Q}$ is Galois. Its Galois group is a semidirect product

$$\mathrm{Gal}(\mathcal{L}_q/\mathbb{Q}) \cong W_q^\vee \rtimes \mathbb{F}_q^\times, \qquad W_q^\vee = \mathrm{Hom}_{\mathbb{F}_q}(W_q, \mathbb{F}_q), \tag{5.1}$$

where $\mathbb{F}_q^\times$ scales the Kummer character. Primes $\ell \equiv 1 \pmod q$ have trivial cyclotomic component, and the property $Q_{M,A}(\chi) \neq 0$ is stable under scaling $\chi \mapsto c\chi$.

Theorem 5.1 (Exact Chebotarev density formula)**.** *Suppose the reduction of $Q_{M,A}$ modulo q is nonzero on $W_q^\vee$. Then the set*

$$\mathscr{D}_q(M,A) = \{\ell : \ell \equiv 1 \pmod q,\ \ell \nmid 6MAq,\ \Delta_{\ell,q}(M,A) \neq 0\}$$

has natural density

$$\operatorname{dens} \mathscr{D}_q(M,A) = \frac{\#\{\chi \in W_q^\vee : Q_{M,A}(\chi) \neq 0\}}{(q-1)q^{r_q}}. \tag{5.2}$$

Relative to the primes $\ell \equiv 1 \pmod q$, its density is

$$\operatorname{dens}_{\ell\equiv 1(q)} \mathscr{D}_q(M,A) = \frac{\#\{\chi \in W_q^\vee : Q_{M,A}(\chi) \neq 0\}}{q^{r_q}}. \tag{5.3}$$

Proof. Let $S_q = \{\chi \in W_q^\vee : Q_{M,A}(\chi) \neq 0\}$. Because $Q(c\chi) = c^2 Q(\chi)$, S_q is stable under the conjugation action of $\mathbb{F}_q^\times$ in (5.1); it is therefore a union of conjugacy classes in $\mathrm{Gal}(\mathcal{L}_q/\mathbb{Q})$. Chebotarev's density theorem [7, 10] assigns it density $\#S_q/\#\mathrm{Gal}(\mathcal{L}_q/\mathbb{Q}) = \#S_q/((q-1)q^{r_q})$. The cyclotomic splitting condition $\ell \equiv 1 \pmod q$ itself has density $1/(q-1)$, giving (5.3). ☐

The required finite-field estimate is sharp in the worst case.

Lemma 5.2 (Universal zero bound for a nonzero quadratic form)**.** *Let q be odd, let $r \geq 2$, and let Q be a nonzero homogeneous quadratic polynomial on $\mathbb{F}_q^r$. Then*

$$\#\{x \in \mathbb{F}_q^r : Q(x) = 0\} \leq (2q-1)q^{r-2}. \tag{5.4}$$

Equivalently,

$$\frac{\#\{x : Q(x) \neq 0\}}{q^r} \geq \left(1 - \frac{1}{q}\right)^2. \tag{5.5}$$

Equality in (5.4) occurs for a split rank-two form such as x_1x_2.

Proof. Pass to the nondegenerate quotient by the radical and let ρ be the rank. For $\rho = 1$, the zero set has q^{r-1} points. For $\rho = 2$, an anisotropic form has q^{r-2} zeros, while a split form has $(2q-1)q^{r-2}$ zeros. For odd $\rho \geq 3$, the standard count is q^{r-1}. For even $\rho = 2m \geq 4$, it is

$$q^{r-1} + \varepsilon(q-1)q^{r-m-1}, \qquad \varepsilon \in \{-1, 1\},$$

which is no larger than the split rank-two count. This proves the bound. Subtracting from q^r yields (5.5). ☐

Theorem 5.3 (Universal torsion transversality)**.** *Let (M,A) arise from a least hypothetical counterexample. For every odd prime q,*

$$\operatorname{dens}_{\ell\equiv 1(q)} \mathscr{D}_q(M,A) \geq \left(1 - \frac{1}{q}\right)^2, \tag{5.6}$$

$$\operatorname{dens} \mathscr{D}_q(M,A) \geq \frac{q-1}{q^2}. \tag{5.7}$$

In particular, for cubic residue characters,

$$\operatorname{dens}_{\ell\equiv 1(3)} \mathscr{D}_3(M,A) \geq \frac{4}{9}, \tag{5.8}$$

$$\operatorname{dens} \mathscr{D}_3(M,A) \geq \frac{2}{9}. \tag{5.9}$$

Proof. By Corollary 3.2, the integral polynomial remains nonzero modulo q. It belongs to $\mathrm{Sym}^2(W_q)$, and the inclusion $W_q \hookrightarrow \overline{\Gamma}_q$ induces an injection on symmetric squares. Because q is odd, a nonzero symmetric tensor defines a nonzero quadratic polynomial on $W_q^\vee$. Apply Lemma 5.2 to (5.3); division by $q-1$ gives the absolute density. □

Cubic Chebotarev certificate

A least counterexample would be locally rank two in cubic power-residue characters at no less than 2/9 of all rational primes. This lower bound is unconditional and uniform in the size and factorization of M and A.

5.1 A local characterization of standard output pairs

The preceding argument applies to any positive integer pair, with the coefficient content measuring the exceptional torsion levels.

Theorem 5.4 (Torsion-profile dichotomy). *Let $M, A \in \mathbb{N}$, and suppose $Q_{M,A} \neq 0$. Put $c = \mathrm{cont}(Q_{M,A})$. For every odd prime q:*

1. *if $q \mid c$, then $\Delta_{\ell,q}(M,A) = 0$ for every eligible ℓ;*
2. *if $q \nmid c$, then the detecting set has the positive density in (5.2), and in particular satisfies (5.7).*

Consequently the following are equivalent:

(a) $M = 2^n$ and $A = 3^n$ for some $n \geq 0$;

(b) the determinant vanishes at every eligible prime for every odd q;

(c) there are infinitely many odd q for which the determinant vanishes on a density-one subset of the primes $\ell \equiv 1 \pmod q$.

Proof. Reduction modulo q is zero exactly when q divides every coefficient, which is exactly $q \mid c$. Otherwise Theorem 5.1 and lemma 5.2 apply. If (c) holds for a nonzero Q, infinitely many primes would divide its nonzero content, impossible. Hence $Q = 0$, and Proposition 3.4 gives (a). The other implications are immediate. □

Remark 5.5. This is a genuine local characterization of the integer-exponent ray. It does not use the real equation (3.2); rather, it says that a nonstandard integer pair cannot hide from all odd torsion characters. The Alaoglu–Erdős difficulty is to make the real rank-one relation interact with this local visibility.

6 Rank-stratified exact densities

The universal lower bound deliberately allows every possible rank drop modulo q. The rank theorem inherited from *Unified Report A* gives more for all but finitely many q.

Let $\rho \in \{3,4\}$ be the rational rank of the symmetric tensor $\tau_{M,A}$. Every nonzero minor witnessing this rank is an integer, so only finitely many primes divide all such minors. Away from those primes, reduction preserves the rank and the multiplicative span.

Theorem 6.1 (Exact generic torsion densities). *For all but finitely many odd primes q, the following alternatives hold.*

1. *If $\rho = 3$, then $r_q = 3$ and*

$$\mathrm{dens}_{\ell\equiv 1(q)}\, \mathscr{D}_q(M,A) = 1 - \frac{1}{q}, \qquad \mathrm{dens}\, \mathscr{D}_q(M,A) = \frac{1}{q}. \tag{6.1}$$

2. If $\rho = 4$, then $r_q = 4$, the reduced form is split hyperbolic, and

$$\operatorname{dens}_{\ell\equiv 1(q)} \mathscr{D}_q(M,A) = 1 - \frac{1}{q} - \frac{q-1}{q^3}, \tag{6.2}$$

$$\operatorname{dens} \mathscr{D}_q(M,A) = \frac{q^3 - q^2 - q + 1}{(q-1)q^3}. \tag{6.3}$$

In either branch, the relative detecting density tends to one as $q \to \infty$.

Proof. For the rank-three branch, the finite quadratic form has a nondegenerate ternary quotient. Every nondegenerate quadratic form in an odd number of variables has exactly q^{r_q-1} zeros, hence nonzero proportion $1 - 1/q$.

For the rank-four branch, the four defining linear forms remain independent and Q is equivalent to $z_1z_2 - z_3z_4$. A split nondegenerate four-variable form has

$$q^3 + (q-1)q$$

zeros. Division by q^4, followed by Theorem 5.1, gives (6.2) and (6.3). $\square$

At a nonexceptional cubic level this gives the more precise table

Generic rational branch	relative density in $\ell \equiv 1 \pmod 3$	absolute density
rank three	2/3	1/3
rank four	16/27	8/27

The robust theorem (5.9) uses only 2/9, because $q = 3$ may be one of the finitely many rank-drop primes.

Remark 6.2 (Why taking q large does not itself prove anything). The relative detecting density approaches one, but the ambient congruence class $\ell \equiv 1 \pmod q$ has density $1/(q-1)$. The absolute density is therefore of order $1/q$. No contradiction follows by letting q grow; one needs a theorem that transfers the real period zero into systematic finite determinant zeros.

7 Characteristic two: Boolean Kummer transversality

The prime two is not covered by the odd-character quadratic-form count, but it gives a slightly stronger absolute density. Put

$$W_2 = \operatorname{span}_{\mathbb{F}_2}\{[2],[3],[M],[A]\} \subset \mathbb{Q}_{>0}^\times/(\mathbb{Q}_{>0}^\times)^2, \qquad r_2 = \dim_{\mathbb{F}_2} W_2,$$

and

$$\mathcal{L}_2 = \mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{M},\sqrt{A}). \tag{7.1}$$

Then $\operatorname{Gal}(\mathcal{L}_2/\mathbb{Q}) \cong W_2^\vee$ and $[\mathcal{L}_2 : \mathbb{Q}] = 2^{r_2} \le 16$.

For an odd prime $\ell \nmid 6MA$, define the additive quadratic character by

$$\left(\frac{u}{\ell}\right) = (-1)^{\chi_{\ell,2}(u)}, \qquad \chi_{\ell,2}(u) \in \mathbb{F}_2,$$

and set

$$\Delta_{\ell,2}(M,A) = \chi_{\ell,2}(M)\chi_{\ell,2}(3) + \chi_{\ell,2}(A)\chi_{\ell,2}(2). \tag{7.2}$$

The plus sign is the reduction of the minus sign in (4.5). As before, $\Delta_{\ell,2} = Q_{M,A}(\chi_{\ell,2})$, and Chebotarev gives

$$\operatorname{dens}\{\ell : \ell \nmid 6MA,\ \Delta_{\ell,2} \ne 0\} = \frac{\#\{\chi \in W_2^\vee : Q_{M,A}(\chi) \ne 0\}}{2^{r_2}}. \tag{7.3}$$

Lemma 7.1 (Boolean degree-two support bound)**.** *Let P be a nonzero homogeneous quadratic polynomial over $\mathbb{F}_2$ on an r-dimensional vector space, with $r \geq 2$. Then*

$$\#\{x \in \mathbb{F}_2^r : P(x) \neq 0\} \geq 2^{r-2}. \tag{7.4}$$

The bound is sharp, for example for $P = x_1x_2$.

Proof. Choose coordinates and replace every x_i^2 by x_i. This gives the unique multilinear polynomial defining the same function on the Boolean cube. It is still nonzero: the square coefficients become distinct linear coefficients, while the mixed coefficients remain distinct quadratic coefficients.

More generally, a nonzero multilinear polynomial of degree at most d on $\mathbb{F}_2^r$ has Hamming weight at least 2^{r-d}. Induct on r, writing $P = P_0 + x_rP_1$. If $P_1 = 0$, the weight doubles from dimension $r-1$. If $P_1 \neq 0$, then coordinatewise $\mathrm{wt}(P_0) + \mathrm{wt}(P_0 + P_1) \geq \mathrm{wt}(P_1)$, and induction at degree $d-1$ gives the same lower bound. Taking $d = 2$ proves (7.4). $\square$

Theorem 7.2 (Quadratic-character transversality)**.** *For a least hypothetical counterexample,*

$$\mathrm{dens}\{\ell : \ell \nmid 6MA,\ \Delta_{\ell,2}(M, A) \neq 0\} \geq \frac{1}{4}. \tag{7.5}$$

Thus at least one quarter of all rational primes detect nonstandardness through the four Legendre-symbol bits of $(2, 3, M, A)$.

Proof. Primitivity of $Q_{M,A}$ keeps its reduction modulo two nonzero. The tensor lies in $\mathrm{Sym}^2(W_2)$; hence its restriction is a nonzero homogeneous quadratic polynomial on $W_2^\vee$. In characteristic two a nonzero homogeneous quadratic remains a nonzero function on the Boolean cube by the first paragraph of Lemma 7.1. Apply (7.3) and Lemma 7.1. $\square$

Strongest uniform absolute-density certificate

A least counterexample would be detected by quadratic characters at density at least 1/4 of all primes, and by cubic characters at density at least 2/9. The quadratic level gives the larger absolute density and a smaller field; the cubic level gives the sharper transfer threshold, because its zero density is at most 5/9 inside one congruence class.

8 Effective detecting primes

The quadratic and cubic levels both have fixed degree. Along with (7.1), set

$$\mathcal{L}_3 = \mathbb{Q}(\zeta_3, 2^{1/3}, 3^{1/3}, M^{1/3}, A^{1/3}). \tag{8.1}$$

Then

$$[\mathcal{L}_2 : \mathbb{Q}] \leq 16, \qquad [\mathcal{L}_3 : \mathbb{Q}] \leq 2 \cdot 3^4 = 162. \tag{8.2}$$

Only primes dividing $6MA$ can ramify. The discriminants of the defining radical polynomials and the standard compositum inequality [7, 9] therefore give absolute effective constants $C_{0,2}, C_{0,3}$ such that

$$\log|D_{\mathcal{L}_j}| \leq C_{0,j} \log(6MA), \qquad j \in \{2, 3\}. \tag{8.3}$$

The constants are independent of the number of prime factors and of the exponent vectors of M and A.

Theorem 8.1 (Effective low-torsion witnesses)**.** *For a least hypothetical counterexample there are eligible detecting primes* ℓ_2, ℓ_3 *with*

$$\Delta_{\ell_2,2}(M,A) \neq 0, \qquad \Delta_{\ell_3,3}(M,A) \neq 0.$$

More precisely, for each $j \in \{2,3\}$*:*

1. *assuming GRH for the Dedekind zeta function of* $\mathcal{L}_j$*,*
$$\ell_j \ll (\log(6MA))^2, \tag{8.4}$$
with an absolute effective implied constant;
2. *unconditionally,*
$$\ell_j \le (6MA)^{C_j} \tag{8.5}$$
for an absolute effective constant C_j*.*

Here $\ell_2 \nmid 6MA$*, while* $\ell_3 \equiv 1 \pmod 3$ *and* $\ell_3 \nmid 6MA$*.*

Proof. At level two, Theorem 7.2 supplies a nonempty subset of $\mathrm{Gal}(\mathcal{L}_2/\mathbb{Q})$ on which $Q \neq 0$. At level three, Theorem 5.3 supplies a Kummer character $\chi \in W_3^\vee$ with $Q_{M,A}(\chi) \neq 0$; its orbit under $\mathbb{F}_3^\times$ is a nonempty conjugacy-stable subset of $\mathrm{Gal}(\mathcal{L}_3/\mathbb{Q})$. Effective Chebotarev [11, 12, 13, 14] applies to either subset. In the cubic case its cyclotomic component is trivial, so the selected prime is $1 \pmod 3$.

Let S_0 be the finite set of primes dividing $6MA$. Use the standard finite-exception variant of effective Chebotarev. In a weighted prime count, deleting S_0 costs at most $\sum_{p \in S_0} \log p \le \log(6MA)$. Under GRH, a first prime outside S_0 in the chosen nonempty conjugacy-stable subset is therefore bounded by a constant times

$$(\log|D_{\mathcal{L}_j}| + [\mathcal{L}_j : \mathbb{Q}] + \log(6MA))^2.$$

Insert (8.2)–(8.3). The unconditional effective counting theorem, with the same finite subtraction, gives a fixed power of $|D_{\mathcal{L}_j}|\,6MA$; another use of (8.3) gives (8.5). The selected primes lie outside S_0, so they are eligible. ☐

Interpretation

Under GRH, any explicit least counterexample would carry two polylogarithmic-size local certificates that it is not on the integer-exponent ray. The quadratic certificate needs only four Legendre symbols in a field of degree at most 16. The cubic certificate needs one cubic root of unity modulo ℓ and four exponentiations in a field of degree at most 162. Neither certificate contradicts the pair's real logarithmic relation.

Remark 8.2 (Uniformity in support)*.* A naive radical extension adjoining a square or cube root of every prime factor of MA would have degree growing exponentially with the support. The fields $\mathcal{L}_2$ and $\mathcal{L}_3$ adjoin only the four radicands $2, 3, M, A$; their degrees are uniformly bounded by 16 and 162. This fixed-degree compression is essential for (8.4).

9 The exact missing bridge: from the real place to cubic density

The Chebotarev theorem determines the finite distribution from the algebraic classes of $2, 3, M, A$. The real counterexample equation is

$$\log M \log 3 - \log A \log 2 = 0. \tag{9.1}$$

There is no formal reduction of the positive real logarithms modulo q, and global reciprocity does not identify the real magnitude character with a finite Kummer character. A proof must therefore supply a genuinely new transfer theorem.

Criterion 9.1 (Cubic density-threshold criterion)**.** *The Alaoglu–Erdős conjecture follows from the following statement:*

> *Whenever positive integers M, A satisfy (9.1) and arise as the least nonintegral output pair, the determinant $\Delta_{\ell,3}(M, A)$ vanishes on a subset of the primes $\ell \equiv 1 \pmod 3$ having relative lower density strictly greater than $5/9$.*

Proof. Theorem 5.3 says that the complementary nonvanishing set has relative density at least $4/9$. Hence the zero set has relative density at most $5/9$, contradicting the proposed transfer statement. □

The threshold is robust: it uses no rank preservation modulo three. In a nonexceptional rank-three branch the permitted zero density is exactly $1/3$; in a nonexceptional rank-four branch it is exactly $11/27$. Thus additional control of the cubic rank could lower the required transfer threshold.

Core open problem isolated by this report

Prove any mechanism by which the archimedean period zero (9.1) forces a statistical excess of finite Kummer rank drops. A density greater than $5/9$ at the cubic level is already sufficient. A theorem forcing $\Delta_{\ell,3} = 0$ for all but finitely many eligible primes would be far stronger than necessary.

9.1 Why ordinary continuity cannot supply the transfer

The map $u \mapsto \log u$ at the real place is continuous in the usual topology. The map $u \mapsto \chi_{\ell,q}(u)$ is a torsion quotient of a finite residue group. Real closeness of two positive rationals says nothing about equality of their power-residue characters. Indeed, the computations in Section 11 exhibit pairs whose real determinant is smaller than 10^{-11} while their cubic characters have the generic rank-four density.

Nor is there a missing application of the product formula. The product formula relates absolute values of one algebraic number over all places. Here the real equation is a quadratic relation among four logarithms, whereas $\Delta_{\ell,q}$ is a quadratic function of four independent Kummer characters. Turning one into the other requires an auxiliary algebraic number whose local behavior encodes the symmetric tensor, and no such construction is currently known.

10 A Chebotarev sieve for synchronized integer gaps

The attachment already studies convergents to a hypothetical β and the synchronized integer gaps. The new torsion theorem adds a precise forbidden-prime statement.

Let r/s be a rational approximation to β, with $r \geq 0$ and $s > 0$, and define

$$D_2(r, s) = M^s - 2^r, \qquad D_3(r, s) = A^s - 3^r. \tag{10.1}$$

For a nonintegral β, neither gap is zero when $r/s \neq \beta$.

Proposition 10.1 (Common gap primes lie on the Kummer quadric)**.** *Let q be prime. For $q = 2$, let $\ell \nmid 6MA$ be odd; for odd q, let ℓ be eligible at level q. Suppose*

$$\ell \mid D_2(r, s), \qquad \ell \mid D_3(r, s), \qquad q \nmid s.$$

Then

$$\Delta_{\ell,q}(M, A) = 0. \tag{10.2}$$

Proof. Apply $\chi_{\ell,q}$ to the two congruences $M^s \equiv 2^r \pmod{\ell}$ and $A^s \equiv 3^r \pmod{\ell}$:

$$s\chi(M) = r\chi(2), \qquad s\chi(A) = r\chi(3).$$

Taking the determinant gives $s\Delta_{\ell,q} = 0$. Since $s \neq 0$ in $\mathbb{F}_q$, the claim follows. $\square$

Corollary 10.2 (Quadratic and cubic forbidden-prime sets)**.** *Let*

$$G(r,s) = \gcd\big(D_2(r,s), D_3(r,s)\big).$$

If s is odd, then $G(r,s)$ has no eligible prime divisor in the quadratic detecting set from Theorem 7.2; for a least counterexample this excludes a set of density at least $1/4$. If $s \not\equiv 0 \pmod{3}$, then $G(r,s)$ has no prime divisor in $\mathscr{D}_3(M,A)$, excluding a set of density at least $2/9$.

Continued-fraction denominators have $\gcd(s_n, s_{n+1}) = 1$, so infinitely many are odd and infinitely many are not divisible by three. Thus the two exclusions apply along infinite subsequences of the best real approximants.

Why the sieve still stops short

An individual integer can be arbitrarily large while all its prime factors lie in the complement of a fixed positive-density set. Consequently the exclusion of the quadratic or cubic detecting sets does not upper-bound $G(r,s)$ strongly enough to repair the exponential denominator loss diagnosed in *Unified Report A*. A successful argument would need an additional distribution theorem for prime divisors of the synchronized gcd sequence, or a lower bound forcing that gcd to sample the detecting Chebotarev set.

The proposition nevertheless sharpens the old statement "real closeness does not impose a common prime divisor." It identifies exactly which Frobenius classes a common divisor is allowed to occupy. This suggests a concrete analytic program: study the sequence $G(r_n, s_n)$ with a Chebotarev-weighted large sieve, rather than estimate only its size.

11 Finite computational stress test

This section is computational evidence only. Its purpose is to test the formulas and to measure how unrelated extreme real closeness can be from cubic torsion rank.

Let

$$\theta = \log_2 3.$$

For each $2 \le M \le 12000$, the search examined the nearest integers to M^θ, removed exact standard pairs, retained pairs whose $Q_{M,A}$ has content one, and sorted by

$$\delta_\infty(M,A) = |\log_2 M - \log_3 A|\,. \tag{11.1}$$

For each retained pair, it found the first eligible cubic detecting prime and measured the nonvanishing frequency among primes $\ell \le 200000$, $\ell \equiv 1 \pmod{3}$.

Computation 11.1 (Near-miss pairs)**.** The following six primitive pairs have very small real mismatch. The last column is the empirical cubic detection rate; all six have the generic split rank-four prediction $16/27 = 0.592592\ldots$.

M	A	$\delta_\infty(M,A)$	first detecting ℓ	empirical rate
9329	1959024	7.6827×10^{-12}	13	$5399/8986 = 0.600824$
8427	1667418	1.1870×10^{-11}	7	$5344/8988 = 0.594571$
10855	2490703	3.1786×10^{-11}	7	$5305/8987 = 0.590297$
9566	2038489	3.4305×10^{-11}	19	$5293/8985 = 0.589093$
10506	2364980	4.2679×10^{-11}	7	$5326/8986 = 0.592700$
9517	2021964	7.3365×10^{-11}	13	$5380/8985 = 0.598776$

For example,

$$9329 = 19 \cdot 491, \qquad 1959024 = 2^4 \cdot 3 \cdot 40813.$$

At $\ell = 13$, one choice of cubic character gives

$$(\chi(2), \chi(3), \chi(M), \chi(A)) = (1, 1, 0, 1),$$

so $\Delta_{13,3} = 0 \cdot 1 - 1 \cdot 1 = 2 \neq 0$ in $\mathbb{F}_3$.

What the computation establishes

The exact finite-field counts match the Chebotarev prediction, and the empirical prime frequencies are close to 16/27. The experiment also falsifies a tempting continuity heuristic: a real determinant of size 10^{-11} need not produce even a mild bias toward cubic rank drop. It says nothing about whether an exact real zero can occur.

12 Milnor K_2, tame symbols, and the parity barrier

The Kummer determinant invites a reciprocity-theoretic attack. This section carries that idea to its exact endpoint. It yields a valid congruence sieve but also exposes a structural sign obstruction.

12.1 Symmetric versus alternating degree two

In additive prime-exponent notation, the target tensor is

$$\tau_{\mathrm{sym}} = \nu(M) \odot e_3 - \nu(A) \odot e_2.$$

If one keeps the displayed signs and antisymmetrizes, one obtains

$$\tau_\wedge^- = \nu(M) \wedge e_3 - \nu(A) \wedge e_2. \tag{12.1}$$

For the standard pair $M = 2^n$, $A = 3^n$, this is

$$ne_2 \wedge e_3 - ne_3 \wedge e_2 = 2ne_2 \wedge e_3,$$

not zero in odd characteristic. The alternating expression that *does* vanish on the standard ray is instead

$$\tau_\wedge^+ = \nu(M) \wedge e_3 + \nu(A) \wedge e_2, \tag{12.2}$$

which has the wrong sign relative to the real determinant. This elementary sign reversal is the core parity barrier.

12.2 The standard-vanishing K_2 class

By Matsumoto's theorem, Milnor and Quillen K_2 agree for $\mathbb{Q}$ [15, 16]. Define

$$\Xi(M, A) = \{M, 3\}\,\{A, 2\} \in K_2^M(\mathbb{Q}). \tag{12.3}$$

Bilinearity and antisymmetry give

$$\Xi(2^n, 3^n) = \{2, 3\}^n \{3, 2\}^n = 1.$$

Thus Ξ detects the exterior expression (12.2), not the symmetric tensor (2.3).

For a rational prime p, use the tame-symbol convention

$$\partial_p\{u, v\} = (-1)^{v_p(u)v_p(v)} \overline{u^{v_p(v)} v^{-v_p(u)}} \in \mathbb{F}_p^\times. \tag{12.4}$$

Write $m_p = v_p(M)$ and $a_p = v_p(A)$.

Proposition 12.1 (Exact tame symbols)**.** *The class* $\Xi(M,A)$ *has the following residues:*

$$\partial_p\Xi(M,A) = 2^{-a_p}3^{-m_p} \pmod p, \qquad p \notin \{2,3\}, \tag{12.5}$$
$$\partial_3\Xi(M,A) = (-1)^{m_3+a_3}\overline{M/3^{m_3}} \in \mathbb{F}_3^\times, \tag{12.6}$$

and $\partial_2\Xi$ *is automatically trivial because* $\mathbb{F}_2^\times = \{1\}$*.*

Proof. For $p \notin \{2,3\}$, formula (12.4) gives $\partial_p\{M,3\} = 3^{-m_p}$ and $\partial_p\{A,2\} = 2^{-a_p}$. Multiplication yields (12.5). At $p = 3$,

$$\partial_3\{M,3\} = (-1)^{m_3}\overline{M/3^{m_3}}, \qquad \partial_3\{A,2\} = 2^{-a_3} = (-1)^{a_3},$$

which proves (12.6). □

Warning 12.2. The symbol Ξ is not known to be trivial for a counterexample. The proposition is a conditional sieve: it records the exact consequences *if* an additional argument can force $\Xi = 1$.

Corollary 12.3 (Congruence consequences of $\Xi = 1$)**.** *If* $\Xi(M,A) = 1$*, then for every* $p \notin \{2,3\}$*,*

$$2^{a_p}3^{m_p} \equiv 1 \pmod p. \tag{12.7}$$

In particular:

- *if* $p \mid M$ *and* $p \nmid A$*, then* $\mathrm{ord}_p(3) \mid m_p$*;*
- *if* $p \mid A$ *and* $p \nmid M$*, then* $\mathrm{ord}_p(2) \mid a_p$*;*
- *if* p *divides both, then their valuation pair lies on the cyclic congruence* $2^{a_p}3^{m_p} = 1$ *in* $\mathbb{F}_p^\times$*.*

There is also a converse at the level of all tame symbols. The localization sequence [16] gives

$$0 \longrightarrow K_2(\mathbb{Z}) \longrightarrow K_2(\mathbb{Q}) \xrightarrow{\oplus_p\partial_p} \bigoplus_p \mathbb{F}_p^\times \longrightarrow 0, \tag{12.8}$$

with $K_2(\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, generated by $\{-1,-1\}$. If every finite tame symbol of Ξ is trivial, then Ξ lies in this order-two kernel. The real Hilbert-symbol signature of a symbol built from positive entries is trivial, whereas $\{-1,-1\}$ has nontrivial real signature. Hence $\Xi = 1$. Thus the congruences above, together with the prime-three condition, exactly characterize the triviality of this positive K_2 class.

12.3 Why norm-residue reciprocity does not encode $Q_{M,A}$

For odd q, the norm-residue isomorphism identifies

$$K_2^M(\mathbb{Q})/q \cong H^2(\mathbb{Q}, \mu_q^{\otimes 2}).$$

The degree-one Kummer cup product is graded-commutative:

$$\alpha \smile \beta = -\beta \smile \alpha.$$

It therefore factors through the exterior square. In contrast, the finite determinant $Q_{M,A}(\chi)$ is obtained by evaluating a *symmetric* quadratic tensor on one character twice. Ordinary degree-two Kummer cohomology has no natural operation that changes the sign in (12.2) back into the sign of τ_{sym} while preserving standard-ray vanishing.

Cohomological parity barrier

The direct ordinary odd-primary Steinberg symbol has the wrong sign for the Alaoglu–Erdős determinant: the standard-vanishing exterior class has the opposite sign, while the same-sign antisymmetrization fails to vanish even for integer exponents. This calculation does not exclude a symmetric refinement or a higher construction.

Characteristic two makes plus and minus coincide, and Section 7 shows that the resulting Boolean quadratic is visible to Frobenius at positive density. This still does not turn the ordinary local Hilbert symbol into $\Delta_{\ell,2}$: for an odd eligible ℓ, all four radicands are ℓ-adic units, and the Hilbert symbol of two units over $\mathbb{Q}_\ell$ is trivial. Thus the diagonal Boolean Frobenius function is not the local invariant of Ξ. Quadratic refinements, Bocksteins, or operations related to the Pontryagin square remain conceivable research directions, but no construction is known that connects them to the positive real logarithmic period.

13 Additional directions tested and their exact stopping points

The new torsion viewpoint suggests several apparently promising shortcuts. None currently closes the conjecture; recording their exact failure modes is part of the progress.

13.1 Sending q to infinity

For generic q, Theorem 6.1 makes almost every character in $W_q^\vee$ detect the tensor. This does not force a contradiction because the rational primes that support q-th power-residue characters satisfy $\ell \equiv 1 \pmod{q}$, a set of density $1/(q-1)$. One would need a uniform theorem coupling many torsion levels at the same rational primes. The fields $\mathcal{L}_q$ vary with q, and no independence statement strong enough for this purpose follows from ordinary Chebotarev alone.

13.2 Reducing the real exponent modulo q

For an integer exponent n, the identities

$$\chi(M) = n\chi(2), \qquad \chi(A) = n\chi(3)$$

make the determinant vanish. It is tempting to replace n by a "residue class of β." There is no canonical map $\mathbb{R} \to \mathbb{F}_q$ compatible with exponentiation on positive rationals. Choosing rational approximants r/s merely replaces this nonexistent residue by the two finite errors

$$s\chi(M) - r\chi(2), \qquad s\chi(A) - r\chi(3),$$

which are unrelated to the smallness of $s\beta - r$ at the real place.

If one deliberately chooses $q \mid s$, both errors become scalar multiples of $\chi(2), \chi(3)$ and their wedge vanishes for the trivial reason that $s = 0$ in $\mathbb{F}_q$. This produces no information about $\Delta_{\ell,q}$. If $q \nmid s$, one recovers the common-gap criterion in Proposition 10.1, but only at primes dividing both integer gaps.

13.3 Global reciprocity without an auxiliary object

Artin reciprocity classifies abelian Frobenius data, and the norm-residue theorem identifies Steinberg symbols with cup products. Neither theorem supplies an algebraic object whose archimedean logarithm is $\log M \log 3 - \log A \log 2$. Reciprocity can compare local invariants of an already constructed class; it does not manufacture the missing symmetric class. The sign computation in Section 12 shows that the most obvious candidate lives in the wrong functor.

13.4 Forcing a small detecting prime unconditionally

Theorem 8.1 gives polynomial bounds in MA without GRH and polylogarithmic bounds under GRH at both the quadratic and cubic levels. Even a uniform numerical bound such as $\ell < 10^6$ would not prove the conjecture: a detecting prime is expected for a counterexample. The needed step is not to find a nonzero determinant, but to contradict it using the real equation.

13.5 Treating the density theorem as probabilistic evidence

The Chebotarev density is exact, not a random-model prediction. Conversely, numerical agreement with that density cannot distinguish a near miss from an exact counterexample. The field $\mathcal{L}_q$ contains no archimedean memory of how small $Q((\log p)_p)$ is. The experiment in Section 11 is therefore best read as a no-go result for continuity heuristics.

Attempt	Exact obstruction
Large odd torsion level q	Relative density rises, but eligible rational primes have density $1/(q-1)$.
A residue class for β	No topology-compatible homomorphism $\mathbb{R} \to \mathbb{F}_q$ exists.
Continued fractions	Real smallness controls gap size, not common Frobenius classes.
Ordinary K_2 reciprocity	Alternating cup products have the wrong sign symmetry.
Effective witness search	A witness confirms nonstandardness; it does not contradict the real relation.
Finite experiments	Exact real equality is an infinite transcendence condition.

14 A focused research program

The new results narrow the next useful work to four programs. They are ordered by how directly they attack Criterion 9.1.

14.1 Program I: Chebotarev-weighted common-divisor theory

For convergents $r_n/s_n \to \beta$, study

$$G_n = \gcd(M^{s_n} - 2^{r_n}, A^{s_n} - 3^{r_n})$$

not merely through $\log G_n$, but through its Frobenius support. For odd s_n, Corollary 10.2 excludes the quadratic detecting classes; for $3 \nmid s_n$, it excludes the cubic detecting classes. A theorem showing that synchronized near-equalities force prime divisors to sample all Kummer conjugacy classes with a positive lower frequency would contradict these restrictions.

A concrete target is the following.

Prime-sampling target

Show that for infinitely many convergents with $3 \nmid s_n$, the common divisor G_n has at least one prime factor in every fixed positive-density Chebotarev set attached to the four radicands $2, 3, M, A$, provided G_n exceeds an explicit height threshold.

This statement is far stronger than standard results on primitive divisors of one recurrence; it concerns a gcd of two multiplicatively independent exponential sequences with synchronized irrational slopes. The formulation makes the required strengthening explicit.

14.2 Program II: a symmetric reciprocity object

The desired finite function is

$$\chi \longmapsto Q_{M,A}(\chi),$$

a quadratic function on H^1, not an alternating bilinear cup product. Possible homes for such information include divided-power algebras, quadratic refinements at the prime two, secondary cohomology operations, or regulators on a higher motivic object. Any candidate must pass three tests:

1. it vanishes identically for $(M, A) = (2^n, 3^n)$;
2. its finite localizations recover $\Delta_{\ell,q}$ or a function with the same zero locus; and
3. its real regulator contains $\log M \log 3 - \log A \log 2$, not an alternating wedge.

The ordinary Steinberg symbol fails the third test by Section 12.

14.3 Program III: a character-sensitive auxiliary determinant

The existing report's auxiliary-function constructions are insensitive to Kummer conjugacy classes. A new determinant could incorporate the idempotent of a detecting orbit in the group algebra

$$\mathbb{Q}(\zeta_q)[\mathrm{Gal}(\mathcal{L}_q/K_q)].$$

The desired architecture is:

1. choose a character χ with $Q(\chi) \neq 0$;
2. project an algebraic auxiliary expression to the corresponding isotypic component;
3. use the exact real period zero to make that component unusually small at one real embedding; and
4. use integrality and product-formula control at the other embeddings to force it to vanish, contradicting the character choice.

The obstacle is simultaneous control of all conjugates. Crude norms multiply too many uncontrolled terms, exactly as in standard radical auxiliary constructions. The group-ring projection is the new ingredient worth testing.

14.4 Program IV: exact formal and computational infrastructure

The finite algebra in this report is particularly suitable for formalization and exhaustive checking. A reusable implementation should:

- represent prime exponent vectors as finitely supported integer maps;
- compute $Q_{M,A}$, its content, and its rank over $\mathbb{Q}$ and $\mathbb{F}_q$;
- verify a power-residue certificate (q, ℓ, ω_ℓ);
- enumerate the finite character space and check the exact density numerator; and
- export a small certificate independent of integer factorization heuristics.

This will not prove the transcendence bridge, but it removes all finite ambiguity from future candidate arguments.

15 Lean formalization blueprint

This section describes a realistic formalization boundary. It does not claim that the new theorems have already been checked by Lean.

15.1 Finite algebra layer

The following components should be routine in Mathlib once the attachment's primitive-output objects are imported.

1. **Prime exponent vectors.** Use a finitely supported function $\mathbb{P} \to_0 \mathbb{Z}$ for $\nu(u)$, together with its reduction modulo q.
2. **Quadratic obstruction.** Rather than implement an infinite polynomial ring, represent
$$Q(z) = \langle \nu(M), z\rangle z(3) - \langle \nu(A), z\rangle z(2)$$
as a function on finitely supported vectors. A finite support containing $\mathrm{Supp}(6MA)$ suffices.
3. **Content theorem.** Expand using (3.4); prove that the gcd of the three coefficient blocks is $\gcd(d, e, |a-b|)$. The central interface with *Unified Report A* is the theorem
`simultaneousCoordinateGCD_eq_one`.
4. **Nonzero reduction.** A primitive finite coefficient family has a nonzero member modulo every prime.
5. **Quadratic zero count.** Either formalize the rank classification over finite fields of odd characteristic, or prove the weaker bound (5.4) by a direct fiber argument and reserve the exact rank formulas for a later file.
6. **Certificate checker.** Verify that $\omega^q = 1$, $\omega \neq 1$, compute the four character values by modular exponentiation, and check the determinant.

Schematic theorem signatures, deliberately avoiding unverified Mathlib API names, are:

```
content_twoStar (normalization : SimultaneousPrimitiveOutput M A) :
  content (twoStarCoefficients M A) = gcd normalization.d
    (gcd normalization.e |normalization.a - normalization.b|)

primitive_twoStar_of_least (h : IsLeastNonintegerSolution beta) :
  Primitive (twoStarCoefficients (2^beta) (3^beta))

quadratic_nonzero_count (q : Nat.Prime) (hq : q != 2)
    (Q : QuadraticForm (ZMod q) W) (hQ : Q != 0) :
  ((Fintype.card W - #{x | Q x = 0}) : Rat) / Fintype.card W
    >= (1 - 1/q)^2
```

These are specifications, not compilable declarations.

15.2 Classical number-theory layer

Kummer theory and Chebotarev are not presently small local lemmas. A practical formal report can separate them as named classical inputs:

- Kummer duality for a finite-dimensional subgroup of $K^\times/(K^\times)^q$;
- the Frobenius/power-residue-symbol correspondence;
- Chebotarev density for a conjugacy-stable finite subset; and
- effective Chebotarev under GRH.

A first formal milestone can prove the finite statement

$$Q \neq 0 \quad \Longrightarrow \quad \#\{\chi : Q(\chi) \neq 0\} \geq (q-1)^2 q^{r-2},$$

and then state the density theorem conditionally on an abstract equidistribution axiom. This would keep the machine-checked and classical layers visibly distinct.

15.3 Certificate format

For $q = 3$, a compact witness consists of

$$(M, A, \ell, \omega; c_2, c_3, c_M, c_A),$$

with checks

$$\begin{aligned}
&\ell \text{ prime}, \quad \ell \equiv 1 \pmod 3, \quad \ell \nmid 6MA, \\
&\omega^3 \equiv 1 \pmod \ell, \quad \omega \not\equiv 1 \pmod \ell, \\
&u^{(\ell-1)/3} \equiv \omega^{c_u} \pmod \ell \quad (u = 2, 3, M, A), \\
&c_M c_3 - c_A c_2 \not\equiv 0 \pmod 3.
\end{aligned}$$

No discrete-log search is needed in the checker: the exponents $c_u \in \{0, 1, 2\}$ are part of the certificate.

16 Audit of the claimed August 2026 breakthroughs

The surrounding claims were checked only to establish context; they are not mathematical inputs to any theorem above.

1. **Jacobian conjecture.** A public 2026 preprint by Shuhong Gao gives a self-contained account of counterexamples in every dimension greater than two, attributing the initial three-dimensional counterexample to Levent Alpöge and the AI-assisted discovery. An independent Lean 4 verification of the explicit three-dimensional map is publicly archived. The plane Jacobian conjecture remains outside that result [17, 18].

2. **Elliptic curve rank at least 30.** ICARM's Elliptic Curve Rank Leaderboard lists curve #273 with 30 explicit independent rational points and hence a certified lower bound $\operatorname{rank} E(\mathbb{Q}) \geq 30$, submitted on 20 August 2026. The page's commentary attributes the search to Claude with Levent Alpöge and Ava Howell. "Exactly 30" is stated there only under GRH and BSD; the unconditional public certificate is the lower bound [19].

3. **Hadamard orders below 2000.** Epoch AI reports, provisionally, that a puzzle posted by a team of three humans and Claude encodes matrices for the twelve previously unknown admissible orders below 2000, including 668. OEIS records the same twelve orders as found in 2026. The attribution and degree of independent reproduction remain more provisional than the existence claim [20, 21].

4. **Alaoglu–Erdős.** Targeted searches on 22 August 2026 found no public preprint, formal certificate, or detailed announcement of the private breakthrough described in the prompt. Absence from a search index is not evidence that no private result exists.

The relevant public pages are recorded in the bibliography. The present report was developed independently from the attached mathematical corpus and those public sources; no alleged private argument was available to inspect.

17 Conclusion

The real equation of a hypothetical counterexample annihilates a primitive symmetric quadratic tensor. The main new observation is that the attachment's affine-primitivity gcd is exactly the coefficient content of that tensor. Once this is recognized, Kummer theory makes the tensor visible at every odd torsion level, and Chebotarev forces nonvanishing at a uniform positive density of rational primes.

The two smallest torsion levels are especially clean:

$$\boxed{\text{least counterexample} \quad \Longrightarrow \quad \operatorname{dens}\{\ell : \Delta_{\ell,2} \neq 0\} \geq \frac{1}{4}, \qquad \operatorname{dens}\{\ell : \Delta_{\ell,3} \neq 0\} \geq \frac{2}{9}.}$$

The fields producing these witnesses have degrees at most 16 and 162. Under GRH, their first witnesses are polylogarithmic in MA. These statements are exact and effectively testable.

The same theorem also clarifies why it is not yet a proof. Finite torsion is generically transverse; a counterexample would not look locally singular. What is missing is a theorem that makes the exact real rank-one relation force too many finite rank drops. The robust cubic threshold is now explicit: more than $5/9$ zero density among $\ell \equiv 1 \pmod 3$ would suffice. The quadratic level supplies the stronger absolute certificate density, but its complementary zero threshold $3/4$ is less demanding only as a certificate and more demanding as a transfer theorem.

The algebraic K-theory audit reaches a similarly sharp conclusion. Tame symbols give a valid local congruence sieve, but ordinary cup products are alternating, while the period obstruction is symmetric. The obvious reciprocity class either has the wrong sign or fails to vanish on the standard integer ray. A successful cohomological proof therefore needs a symmetric quadratic refinement, not merely a Steinberg symbol.

Most promising next step

Combine the positive-density detecting set with the synchronized gap sequence $\gcd(M^s - 2^r, A^s - 3^r)$. The new sieve says every common prime divisor lies on a finite Kummer quadric whenever the chosen torsion prime q does not divide s. A prime-distribution theorem forcing these gcds to sample the complementary Chebotarev classes would convert the present transversality theorem into a contradiction.

A Reproducibility code for the cubic experiment

The following compact Python program reproduces the character calculation and empirical density check for a supplied list of pairs. It requires `sympy`. The exhaustive near-miss search used the same routines and checked both nearest integers to M^θ.

```
from itertools import product
from sympy import factorint, primerange, primitive_root

Q = 3
PAIRS = [
    (9329, 1959024), (8427, 1667418), (10855, 2490703),
    (9566, 2038489), (10506, 2364980), (9517, 2021964),
]

def q_character(u: int, ell: int, q: int, primitive: int) -> int:
    omega = pow(primitive, (ell - 1) // q, ell)
    value = pow(u % ell, (ell - 1) // q, ell)
    cur = 1
    for exponent in range(q):
        if cur == value:
            return exponent
```

```
        cur = (cur * omega) % ell
    raise RuntimeError("power-residue value not in mu_q")

def q_polynomial(M: int, A: int, support, z, q: int) -> int:
    fm, fa = factorint(M), factorint(A)
    LM = sum(fm.get(p, 0) * z[i] for i, p in enumerate(support)) % q
    LA = sum(fa.get(p, 0) * z[i] for i, p in enumerate(support)) % q
    X2 = z[support.index(2)]
    X3 = z[support.index(3)]
    return (LM * X3 - LA * X2) % q

for M, A in PAIRS:
    support = sorted(set(factorint(6 * M * A)))

    # Exact finite-character proportion. Restriction to W_q has equal fibers.
    total = q ** len(support)
    nonzero = sum(
        q_polynomial(M, A, support, z, q) != 0
        for z in product(range(q), repeat=len(support))
    )

    eligible = detected = 0
    first = None
    for ell in primerange(2, 200001):
        if ell in support or (ell - 1) % q != 0:
            continue
        primitive = primitive_root(ell)
        c2 = q_character(2, ell, q, primitive)
        c3 = q_character(3, ell, q, primitive)
        cM = q_character(M, ell, q, primitive)
        cA = q_character(A, ell, q, primitive)
        delta = (cM * c3 - cA * c2) % q
        eligible += 1
        if delta:
            detected += 1
            if first is None:
                first = (ell, delta, (c2, c3, cM, cA))

    print(
        M, A,
        f"exact={nonzero}/{total}={nonzero/total:.6f}",
        f"empirical={detected}/{eligible}={detected/eligible:.6f}",
        f"first={first}",
    )
```

For the six pairs in Computation 11.1, the exact finite-character proportion is 16/27. The prime cutoff 200000 gives approximately 8990 eligible primes per pair.

B Proof details for the finite-field counts

For completeness, let Q have rank ρ on an r-dimensional vector space over $\mathbb{F}_q$, q odd. Factoring out the radical contributes a multiplicative factor $q^{r-\rho}$ to the number of zeros. For a nondegenerate form in ρ variables,

$$N_0(Q) = \begin{cases} q^{\rho-1}, & \rho \text{ odd},\\ q^{\rho-1} + \varepsilon(q-1)q^{\rho/2-1}, & \rho \text{ even}, \end{cases}$$

where $\varepsilon = 1$ for the split even-dimensional form and -1 for the nonsplit form. At rank two, the split count is $2q-1$, the maximum among all nonzero quadratic forms. Multiplying by the radical factor proves (5.4).

For the rank-four hyperbolic form $z_1z_2 - z_3z_4$, the formula gives

$$N_0 = q^3 + (q-1)q, \qquad 1 - \frac{N_0}{q^4} = 1 - \frac{1}{q} - \frac{q-1}{q^3}.$$

For $q = 3$, this is $16/27$, the exact value used in the computational table.

C Source and dependency ledger

Statement	Status	Dependencies
Least-output normalization and gcd one	inherited	Attached *Unified Report A*, simultaneous primitive output normalization
Rank three/four of the symmetric tensor	inherited/classical	Attached *Unified Report A*; Baker linear independence
Content formula (3.3)	proved here	Elementary coefficient comparison
Kummer rank injection	proved here	Norm argument; standard Kummer theorem
Exact density (5.2)	proved here from classical input	Kummer–Frobenius correspondence; Chebotarev
Universal odd-primary density (5.7)	proved here	Content formula; finite quadratic zero count; Chebotarev
Quadratic density (7.5)	proved here	Content formula; Boolean minimum distance; multiquadratic Chebotarev
Generic exact densities	proved here	Inherited rank classification; finite-field counts
GRH low-torsion witness bounds	conditional	Effective Chebotarev under GRH; fixed-degree discriminant bounds
Unconditional witness bounds	proved from classical effective theorem	Unconditional least-prime Chebotarev bound
Gap-prime sieve	proved here	Elementary character calculation
Tame-symbol formulas	proved here	Matsumoto theorem; localization sequence for $K_2(\mathbb{Q})$
Parity barrier	proved here as a structural no-go	Graded commutativity of cup products
Near-miss table	computation	Python/Sympy exact modular arithmetic
Cubic density transfer	open target	Any archimedean-to-Kummer rank-drop theorem

References

[1] L. Alaoglu and P. Erdős, *On highly composite and similar numbers*, Transactions of the American Mathematical Society **56** (1944), 448–469. https://doi.org/10.1090/S0002-9947-1944-0011087-2.

[2] *The Alaoglu–Erdős Integer-Exponent Conjecture: Machine-Checked Partial Results, Exact Conditional Structure, Determinant Methods, and the Remaining Barrier*, attached unified manuscript, accessed 22 August 2026.

[3] A. Baker, *Linear forms in the logarithms of algebraic numbers*, Mathematika **13** (1966), 204–216.

[4] S. Lang, *Introduction to Transcendental Numbers*, Addison–Wesley, 1966.

[5] K. Ramachandra, *Contributions to the theory of transcendental numbers, I–II*, Acta Arithmetica **14** (1967/68), 65–88 and 73–88.

[6] J. S. Milne, *Fields and Galois Theory*, version 5.10, 2022, https://www.jmilne.org/math/CourseNotes/ft.html.

[7] J. S. Milne, *Algebraic Number Theory*, version 3.08, 2020, https://www.jmilne.org/math/CourseNotes/ant.html.

[8] J. S. Milne, *Class Field Theory*, version 4.03, 2020, https://www.jmilne.org/math/CourseNotes/cft.html.

[9] J. Neukirch, *Algebraic Number Theory*, Springer, 1999.

[10] J.-P. Serre, *Lectures on $N_X(p)$*, Research Notes in Mathematics 11, CRC Press, 2012.

[11] J. C. Lagarias and A. M. Odlyzko, *Effective versions of the Chebotarev density theorem*, in *Algebraic Number Fields*, Academic Press, 1977, 409–464.

[12] L. Grenié and G. Molteni, *An effective Chebotarev density theorem under GRH*, Journal of Number Theory **200** (2019), 441–485; https://arxiv.org/abs/1709.07609.

[13] J. Thorner and A. Zaman, *A unified and improved Chebotarev density theorem*, Algebra & Number Theory **13** (2019), 1039–1068; https://arxiv.org/abs/1803.02823.

[14] H. Kadiri and P.-J. Wong, *Primes in the Chebotarev density theorem for all number fields*, Journal of Number Theory **241** (2022), 550–582; https://arxiv.org/abs/2105.14181.

[15] J. Milnor, *Introduction to Algebraic K-Theory*, Annals of Mathematics Studies 72, Princeton University Press, 1971.

[16] C. A. Weibel, *The K-Book: An Introduction to Algebraic K-Theory*, Graduate Studies in Mathematics 145, American Mathematical Society, 2013.

[17] S. Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222, 2026; https://arxiv.org/abs/2608.00222.

[18] P. Nogueira Grossi, *An Independent Lean 4 Verification of the Alpöge–Fable Counterexample*, Zenodo, 23 July 2026; https://doi.org/10.5281/zenodo.21514514.

[19] Institute for Computer-Aided Reasoning in Mathematics, *Elliptic Curve Rank Leaderboard, curve #273*, accessed 22 August 2026; https://elliptic-rank.icarm.cloud/curve/273.

[20] Epoch AI, *Hadamard Matrix of Order 668*, provisional solution update, accessed 22 August 2026; https://epoch.ai/frontiermath/open-problems/hadamard.

[21] OEIS Foundation, *A019442: Numbers m such that a Hadamard matrix of order m exists*, 2026 update; https://oeis.org/A019442.